Wildchrokie ms

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Introduction

Methods

Common garden setup

We collected seeds from four provenances across a 3.5° latitudinal gradient (Figure 1 and Table 1) in 2014-2015, germinated them following standard germination protocols and grew them to seedlings. In the spring of 2017, we planted them outside in the garden located at the Weld Hill Research Building at the Arnold Arboretum in Boston MA (42.30°N, 71.13°W). Plots were regularly weeded and watered throughout the duration of the study and were pruned in the fall of 2020. We initially collected seeds of 18 species of northeastern North American trees and shrubs. We exclude the shrubs, and the five species with low survivorship in the analyses. Thus, we have four species in the *Betulaceae* family (Table 1).

Phenological monitoring and sample collection

For the years 2018-2020, we monitored leaf phenology for all individuals in the common garden once or twice per week from February to December (Figure S1). See (Buonuito, unpublished) for more details about the phenological monitoring. In the spring of 2023, we collected cross-sections for 78 individual trees, cutting the trunk just above the root collar of most trees. We also collected tree cores for 43 individuals using standard dendrochonological methods and stored them in modified plastic straws to allow aeration. Both the cross-sections and cores were left to dry at ambient temperature for three months.

Sample processing, imaging and measuring

We mounted the cores on wooden mounts, and sanded both the cores and cross-sections using progressively finer sandpaper grits: 150, 300, 400, 600, 800, 1000. We scanned the cores and cross-sections at a resolution of 6250 dpi, with a high-resolution treering scanner (Fong, unpublished), we then used the digitized images to measure the tree ring widths with ImageJ (Schneider *et al.*, 2012). For individuals for which we had cross-sections, we measured ring widths at three standard locations on the cross-section, at 120° from each other. We averaged the three measurements, to yield one ring width per year, per individual. For the 8 individuals for which we only had cores, we measured only one ring width for that sample. When we had cross-sections and cores for an invidividual, we discarded the measurement from the core.

We performed visual cross dating, but did not perform statistical crossdating because of the short chronologies that limit the capacity of these analyses (Raden *et al.*, 2020). Given the short lifespan of the trees in our dataset, we did not detrend the ring widths. However, we included a year intercept that accounts for other factors that could affect growth, such as age allometry.

Data analyses

To understand how growth shifted with season lengths we used linear Bayesian hierarchical models. With these models, we estimated ring width as a function of four different season predictors. First, the thermal

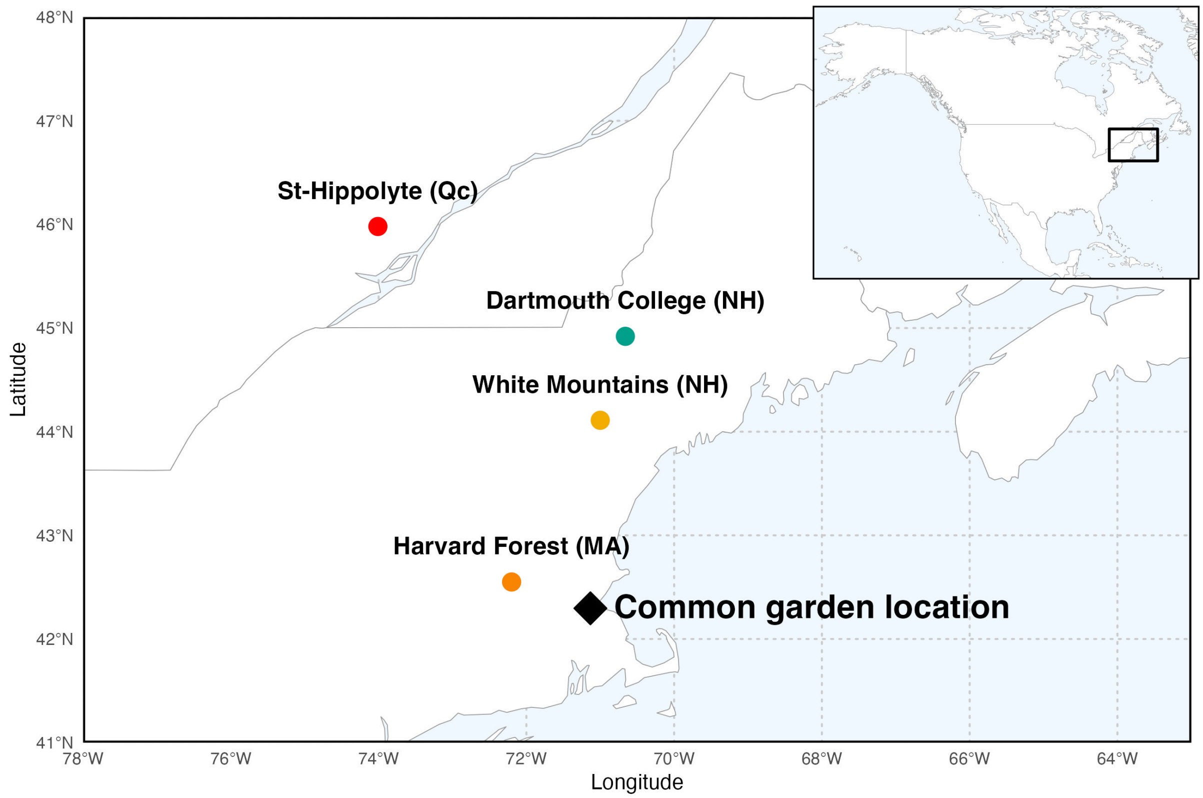

Figure 1: Provenance are depecticted as circles, while the star denotes the location of the common garden the seeds accross a latitudinal gradient. The common garden location is where the study took place at the Weld Research Building (Boston, MA).

season represents the number of growing degree days (GDD) calculated as the mean daily temperature above 5°C accumulated between leafout and budset, calculated for each tree per year. Second, the calendar season or growing season length (GSL) is the budset date subtracted by the leafout date. Finally, to better understand trees responses to the calendar season, we used leafout as the start of the season (SOS) and budset for the end of season (EOS) as separate predictors that represent the boundaries of the growing season.

$$
\begin{aligned}
ln(ringwidth)_i &\sim N(\mu, \sigma_y) \\
\mu &= \alpha + \alpha_{\text{species}} + \alpha_{\text{site}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} + \beta_{species} \times \text{season predictor} \\
\alpha &\sim N(1, 3) \\
\alpha_{\text{species}} &\sim N(0, 6) \\
\alpha_{\text{site}} &\sim N(0, \sigma_{\alpha\,\text{site}}) \\
\alpha_{\text{treeid}} &\sim N(0, \sigma_{\alpha\,\text{treeid}}) \\
\beta_{\text{species}} &\sim N(0, 0.5) \\
\sigma_{\alpha\,\text{site}} &\sim N(0, 1) \\
\sigma_{\alpha\,\text{treeid}} &\sim N(0, 1) \\
\sigma_y &\sim N(0, 3)
\end{aligned}
$$

where i is each observation of the dataset. Intercepts on species ($\alpha_{species}$) and year (α_{year}) and slope on species ($\beta_{species}$) were not pooled (fixed effects), and we partially pooled (random effects) site (α_{site}) and treeid (α_{treeid}) to account for the similarity of the measurements within these groups. We ran four chains with 2000 warmup iterations each, which we discarded, and 2000 sampling iterations, which we kept for posterior distribution estimates. The models did not have any divergent transitions, had a $\hat{R}$ below 1.01, and values of effective sample size (n_{eff}) greater than 10% of the model iterations. We fit ln(ring width) so the data distribution was normally distributed. We wrote our models using Stan (Carpenter *et al.*, 2017) with the rstan package version 2.32.7 (Stan Development Team, 2025) to run the Stan code in R.

To account for differences in temperature across years, we used the daily climate data from the climate station located at Weldhill research station for 2018 and 2019 (`https://arboretum.harvard.edu/research/data-resources/weather/`). Because the data was not available for that station after August 2020, we used the daily climate data from the weather station "Boston (Weld Hill)" of the Network for Environment and Weather Applications (NEWA) for the whole year in 2020 (`https://newa.cornell.edu/all-weather-data-query/`). We calculated daily growing degree days (GDD) by subtracting a 5°C threshold from the mean daily temperature.

As a supplementary analyses, we ran the models using basal area increment (BAI), as the response variable. For this, we summed each ring width separately at the three distinct locations on the cross sections yielding three radii from pit to bark. We then calculated the area of each ring by subtracting the radius from the inner ring from the outer ring:

$$\pi \times (\text{Inner radius}^2 - \text{Outer radius}^2)$$

We then averaged across the three BAI for each ring and fit the models with the four different predictors.

Table 1: Provenance coordinates and count per species and provenances. We had a total of 81 individual trees

Provenance	Longitude	Latitude	*A. incana* ($n = 29$)	*B. alleghaniensis* ($n = 20$)	*B. papyrifera* ($n = 7$)	*B. populifolia* ($n = 25$)
Harvard Forest (MA)	-72.2	42.5	7	0	4	7
White Mountains (NH)	-71.0	44.1	6	7	0	9
Dartmouth College (NH)	-70.7	44.9	8	7	2	9
St-Hippolyte (Qc)	-74.0	46.0	8	6	1	0

Results

Summary basic information

Wildchrokie includes four deciduous tree species from four sites with leaf phenology data spaning three years (2018 to 2020) for a total of 239 leafout and 230 budset observations, yielding 227 complete growing seasons (GS) for 81 individuals.
The earliest leafout day of year was 22-Apr (*A. incana* in 2019) and the latest, 05-Jun (also *A. incana*, but in 2020). The earliest budset day of year was 13-Aug (*B. alleghaniensis* in 2018) and the latest, 24-Oct (*A. incana* in 2018)
Mean summer temperature were similar across years with 2019 being the coolest (22.383°C), 2020 the hottest (23.067°C), and 2018 in between (22.85°C). There was no drought for the three years, except for in 2020, where a moderate summer drought ramped up to a severe drought in September. Annual ring widths across years varied from 0.389 mm (*B. alleghaniensis* in 2020) to 13.547 mm (*B. populifolia* in 2018) (Figure **??**).

0.1 Does tree growth increase with longer growing seasons?

Longer growing seasons increased growth for all species, but varied depending on the metric used to measure it. Longer thermal season increased growth for *B. papyrifera* ($\beta = 0.443$, UI_{90} [0.179, 0.703]), *B. populifolia* ($\beta = 0.215$, UI_{90} [0.072, 0.357]), and *B. alleghaniensis* ($\beta = 0.129$, UI_{90} [0.013, 0.243]), but *A. incana* had a weak effect, but trended in the same direction ($\beta = 0.067$, UI_{90} [-0.037, 0.169]) (Figure **??**a and e) (Table**??**).

A. incana grew more with longer calendar seasons ($\beta = 0.05$, UI_{90} [0.004, 0.097]), but of the three species that responded to thermal season, only *B. papyrifera* also grew more with longer calendar seasons ($\beta = 0.184$, UI_{90} [0.051, 0.317]).

B. alleghaniensis ($\beta = 0.033$, UI_{90} [-0.023, 0.089]), and *B. populifolia* ($\beta = 0.054$, UI_{90} [-0.019, 0.127]) trended in the same direction (Figure **??**b and f) (Table **??**).
Results with basal area increment (BAI) for the GDD model were broadly similar to ring width (see Table **??**)

Discussion

References

Carpenter, B., Gelman, A., Hoffman, M.D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P. & Riddell, A. (2017). *Stan* : A Probabilistic Programming Language. *Journal of Statistical Software*, 76.

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Supplemental results

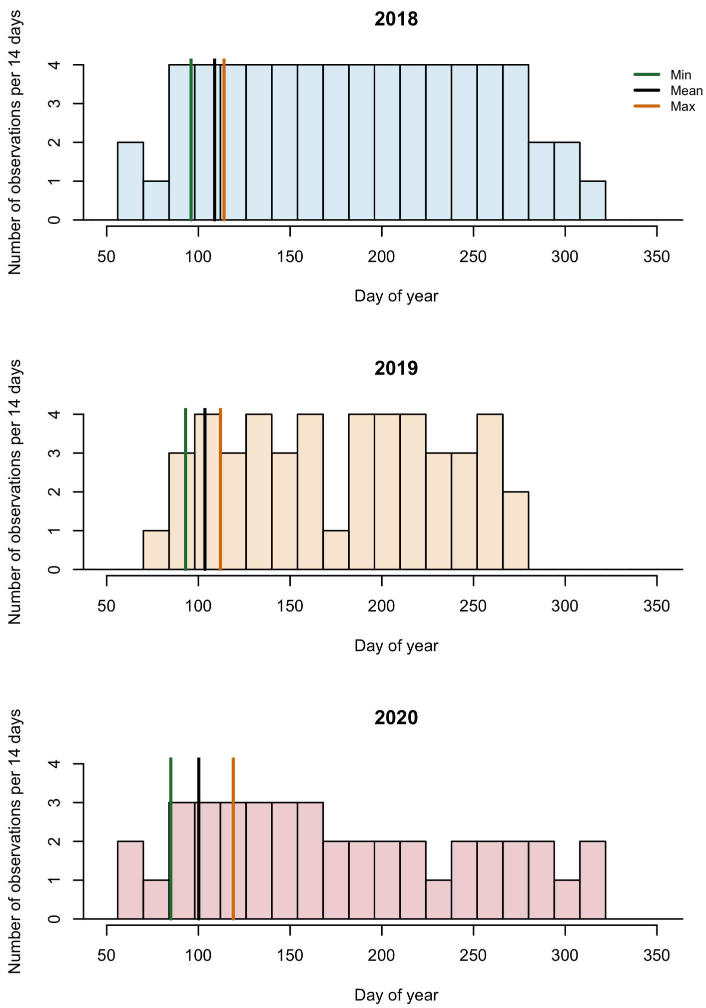

Figure S1: Phenology observatons bi-weekly.